

(OpenAI GPT-4o-mini) that reads the posting text together with the candidate list and selects the tasks it judges to be genuinely required. The procedure yields a panel of 2,748,186 posting-task pairs. Appendix A.14 describes the pipeline in detail, reports the exact prompt and JSON schema, and presents sanity checks on the output.

4.1.2 Main Sample

Our final sample merges all U.S. positions in the Revelio Labs dataset with job postings at the firm level. The resulting dataset covers 281,111 firms that were successfully matched to both employee position data and job postings, and that were actively hiring between January 2021 and March 2025.¹⁸ For these firms, we observe 161,192,969 positions held by 65,470,800 workers dating back to 2015, as well as 198,773,384 job postings since 2021, all with raw text descriptions.¹⁹

4.1.3 Measuring Workforce Dynamics by Seniority

We construct a monthly panel at the firm level. For each firm-by-month, we calculate the number of employees who held a position at the firm that started before and ended after that month, capturing the firm’s workforce size in that period. We repeat this calculation separately for each seniority category, enabling us to track the workforce composition over time. Additionally, we identify monthly inflows and outflows by seniority. For each firm-by-month, we define new hires as workers who began a new position at the firm that month, having most recently worked at another firm or for whom this is their first observed job. Separations are defined as workers whose position at the firm ended in that month and who either moved to a different firm or had no subsequent position listed. Finally, we define promotions as workers who start a new position at the firm after

¹⁸A firm is considered active if it had at least 20 new positions starting during this period.

¹⁹We exclude the top 1 percent of firms with the highest postings-to-hires ratios. Manual inspection indicates that many of these firms are HR intermediaries recruiting on behalf of other employers. In addition, we exclude the approximately 800 largest firms in the Revelio data (ranked by post-2021 new hires), representing about 0.2 percent of the firms in the sample, to avoid the influence of extreme outliers on our estimates. Supplemental Appendix A.19 shows that our main triple-difference results remain robust when these firms are added back.